OpenSourceRail v0.5

Ferrite PM maglev — primary design reader edition

Fixed magnetic modules · compact elevated guidebeams · Rust controls · local manufacture

Provisional digital studies. Complete vehicle, civil and software qualification remains open. Accepted physical releases: 0.

OpenSourceRail v0.5.2 development — station-powered ferrite maglev

docs/open-source-rail-overview.md

Fewer moving parts, local manufacture and compact elevated urban routes.



OpenSourceRail maglev station concept

Illustrative development concept, without physical or manufacturing acceptance.

Station-mounted linear synchronous motors launch and receive the train. Interstation propulsion is unpowered; onboard ferrite PM-biased active support, guidance and passenger services remain independently powered. Removing main traction equipment from the train shifts electrical plant to stations. Support coils/drives, cooling, sensing, protection, doors, HVAC and emergency equipment remain necessary. Contactless running aims to reduce contact noise and wear; actual losses, acoustic performance and maintenance require measurement.

Body, civil works and local capability

Baghdad uses six 17 m bodies. The central tare target is 14.2 t/car, including a 5.2 t equipped structural GFRP body, 3 t ferrite, 30 kWh services storage and the full 1.8 t growth reserve. Four passenger doors per car are the body takeoff. Calculated panel stock is distinct from sized members, joints, complete equipment mass and structural qualification.

The ordinary civil candidate is a 25 m, 0.80 × 1.80 m closed concrete box with 100 mm webs, 150 mm flanges and end infills. Hollow columns/caps and alternative foundation families are compared by quantity. Necessary passive iron, anchors, bearings, connections and final survey remain in scope. Ground-specific selection, prestress/rebar detailing, loads and construction stages remain unqualified.

The research target omits a continuous external walkway. Vertical spiral chutes are the preferred alternative under investigation, without walkway-removal approval. Safe landing, assisted evacuation, deployment, stopping/recovery and fire response must work at every allowed stopping location.



Vertical spiral escape chute concept

Illustrative concept; no qualified railway device or all-stop evacuation claim.

The 99% elevated-mainline objective and straight tangents aim to reach constrained corridors with less surface occupation. Rights, buildings, utilities, access, wildlife, water, curvature and foundation feasibility govern each route.

Local precast/GFRP production, steel fabrication, cartridge assembly, station winding cassettes, interiors, wiring, Rust software and integration can build domestic capability. Prioritise verified British manufacture when selecting UK machinery. A UK distributor or integrator does not establish manufacturing origin. Complete local content remains unknown until the bill and input origins exist.

Operating and cost boundaries

The station operating reference is a jerk-limited 90 s flat 1 km run with 190 s headway. Grade, wind, drag, arrival speed and receiving-brake readiness remain explicit route gates. A committed coasting train keeps its corridor and receiving berth reserved until proved full-consist clearance. Receiver failure, independent stopping/holding and stationary recovery remain design obligations; a distant station motor cannot restart a train in an unpowered span.

A 45 s charge within a 60 s dwell needs 1.20 MW/train before charger losses under the hot-services assumption, including the battery energy used during the remaining dwell. Station electrical cost targets are additional to the incomplete ordinary civil subtotal. Stations, external grid work, complete evacuation, recovery, land, qualification, factory capital and financing need their own scope. No quoted complete railway price or accepted production rate is established.

Inspect and reproduce the current work

Current digital capability	Scope
City planning	266 primary packages using explicit geography/demand and recalculated formation, duty, fleet and costs
Operating study	Jerk-limited launch/coast/receive, assumed magnetic-band overlap, power, receiving heat and charging
Mechanical/civil quantities	Three GFRP cases, magnet-orientation inventories, five beam families, ground sensitivities and a conditional production bottleneck
Controls	Independent Rust support and station demand studies, persistent launch reservations and offline negative controls
Simulation and media	Partial native CAD/images, assembly visibility videos and source receipts; coupled replay remains a stationary single car with a vertical beam
Engineering acceptance	0 accepted physical releases; design, manufacture and operational gates remain open

Ring benchmarks and block-orientation seeds receive no railway lift or mass-saving credit. Moving six-car magnetic/structural dynamics, hardware timing, complete manufacturing geometry and physical acceptance remain open. Visibility videos show final-position parts, without motion, collision or erection-strength simulation.

Use `./osr build` to verify dependencies and rebuild stale implemented studies. `./osr design station`, `./osr design structure` and `./osr design protocol` reproduce their separate evidence. `./osr workbench` opens the review UI; `./osr book` builds the current reader edition. The scheduled runner refreshes implemented evaluators after source changes; it does not invent physical evidence.

Open the Workbench, review calculations, CAD and videos, read implementation and open work, review v0.5.2 development notes, or download the reader book.

Published v0.5.1 remains preserved. This document describes v0.5.2 development on main and does not announce a new release.

OpenSourceRail v0.5 station-powered ferrite-maglev engineering

engineering/ferrite_maglev/README.md

The primary design uses station-mounted linear synchronous motors for launch and receiving braking, with unpowered interstation coasting. Onboard ferrite PM-biased active support/guidance and services storage remain independent. Structural GFRP sandwich bodies, repeated narrow concrete guidebeams and Rust control packages are the leading studies. Accepted engineering and physical releases: zero.

Run and inspect

```
./osr build
./osr design station
./osr design structure
./osr design protocol
./osr design iterate
./osr design check
./osr design verify
./osr city baghdad
./osr sim baghdad --cycles 2
./osr workbench
./osr test
./osr book
cargo test --locked -p osr-maglev-control --all-targets
```

Current report · Workbench · city directory · architecture.

`./osr capabilities` publishes executable sources and scope. `./osr catalogue plan --city baghdad` resolves a selected regeneration, and `./osr catalogue run --city baghdad` rebuilds that city. `./osr readiness` reports source and acceptance gates. `./osr design release-check` remains held while engineering acceptance is open.

The dependency runner verifies implemented native, numerical, control, city and publication prerequisites. An interprocess lock excludes concurrent timer/manual builds. Progress is retained under `build/maglev-workflow/latest.json`. The runner rebuilds stale evidence and holds dependent work after failures; missing evaluators and physical measurements remain additional engineering work.

Current inputs and design records

Concern	Source
Architecture	Infrastructure policy
Vehicle/civil configuration	Configuration, reader, primary inputs
Station motion and electrical plant	Operating model, published trajectory/overlap/power study
GFRP, magnet orientations and civil production	Structure model, study inputs, published quantities
Requirements and allocation	Requirements, systems register
Costs and manufacture	Process model, planning rates
Magnetic/native geometry	Geometry, field controls, native exporter, station-native packet
Station reservation protocol	Offline protocol evidence, persistent Rust interlocking ownership
City geography/service	Each city's `city.json`; compiler
Onboard support control	Rust package, offline host profile

Baghdad uses six 17 m bodies, with 104.5 m assumed overall length pending coupler and gangway design. The central tare target is 14.2 t/car, including a 5.2 t equipped GFRP body, 3 t ferrite, 30 kWh services battery and full 1.8 t growth reserve. Four passenger doors per car are the current body takeoff. Density, laminate/core substitutions, unsized members and body-side/magnetic-carrier ownership are explicit. These budgets are distinct from weighed hardware and a complete manufactured-parts BOM.

The ordinary guidebeam candidate is a 25 m, 0.80 × 1.80 m closed box with 100 mm webs, 150 mm flanges and two 0.4 m end infills. Concrete plus unchanged passive iron is about 43.1 t before outstanding reinforcement/attachment detail mass and rigging. Hollow columns, caps and shallow/single-shaft/pile-group foundations are quantity studies; qualified ground selection remains absent.

Implemented checks and open boundaries

Capability	Implemented scope	Evidence still required
Station trajectory	Jerk-limited launch, zero-thrust coast and receiving brake; energy closure and assumed magnetic-band overlap	Measured drag/magnetic losses, actual grades/wind, useful thrust, voltage/current/thermal limits and robust arrival margins
Station electrical costs	Fixed-group/section alternatives and separate power/charging target-rate rows	Configured equipment, supplier quotations, grid capacity, real duty and complete station scope
Body/civil quantities	GFRP panel mass cases, interface ownership, beam/end-infill quantities, support/foundation screens and target costs	Orthotropic structural analysis, member/joint/prestress/rebar detailing, soils, tolerances and construction-stage acceptance
Magnet orientations	Point-dipole ring reference and finite-block orientation inventories	Nonlinear finite-geometry steel/coil force/loss/demagnetization maps and measured shared-field behavior; no ring-gain credit
Native geometry	Partial panel/interface/civil/station definitions, export/reopening checks, retained images and visibility sequences	Complete production geometry, joints, equipment, drawings and qualified manufacturing tolerances
Rust support control	Module allocation, virtual replay, protection interfaces and support-preserving stop obligations	Actual magnetic/sensor plant, independent protection, physical I/O and measured timing
Launch ownership	Offline corridor/receiving-berth reservation, committed occupancy and full-clearance protocol checks	Authenticated complete-consist sensing, real receiving readiness, independent stopping and qualified system integration
Coupled replay	One stationary rigid car exchanging ideal support forces with a flexible vertical beam	Moving six-car loads, flexible vehicle, lateral/torsional/joint/soil response and correlated faults/ride/stability
Adaptive design	Previous-result-guided vehicle/beam/gain proposals, ancestry, Pareto observations and separate confirmation	Completed parts and qualified evaluators/plant before manufacturing or operating selection
Factory and city delivery	Conditional two-shift bottleneck capacities, duty/fleet, route quantities and process needs	Measured cure/crew/logistics rates, surveys, route recovery, complete economics and accepted instances

The adaptive search, configured in iteration.toml, has explicit evaluation/stagnation budgets and exact-source restart receipts. Failed confirmation disqualifies the exact revision. Published history retains proposals, ancestry, traces and recommendation. Mass/inertia allocations and preference weights remain research assumptions. An analytical recommendation does not automatically change acceptance criteria or reference configuration.

Operation, manufacture and costs

The operating reference is a 90 s flat 1 km trip with 0.8 m/s² acceleration and service deceleration, 0.6 m/s³ jerk and 190 s scheduled headway. The 180 s case has little clearance margin at the assumed 60 s dwell. Grade, wind, drag, arrival speed and receiving-brake availability must be resolved per span. A coasting train already committed to a corridor cannot be stopped by expiring its permission. Persistent ownership and positive tail clearance are required. Receiver faults, independent stopping/holding and a train stranded outside powered zones remain separate design requirements; a distant stator cannot restart it.

At 60 kW/car hot services, a 90 s trip uses 9 kWh/train. If 45 s of a 60 s dwell is available for charging, the remaining 15 s adds 1.5 kWh to battery demand. Restoring 10.5 kWh in 45 s takes 840 kW, plus 360 kW direct services: 1.20 MW/train before charger losses. Battery reserve, repeated charge duty and ageing remain unqualified. The 60 s charging alternative needs a distinct rated interface.

The supplied civil target-rate comparison falls from about USD 4.255 million to USD 3.313 million per two-direction ordinary route-km. It is an incomplete no-walkway scope and the USD 22,000 ordinary-support price is a procurement target. Station propulsion/charging equipment is additional, separately labelled scope. Reserve does not price missing station structures, evacuation, land, factory capital, qualification, recovery, finance or other programme work. Lower mass alone earns no proportional ground or support saving.

City manufacturing records carry material quantities, process routes, equipment needs and construction approaches. Candidate processes include controlled GFRP infusion/RTM and bonding, precast mould/cage/prestress/cure work, reaction-iron and carrier fabrication, station coil winding, cartridge assembly, wiring and tests. The production model takes the minimum of factory, foundation, pier, delivery, erection and final survey capacities. Its 56.25 route-m/planned-day target is not measured output. Demould, connection strength, crane-at-radius, ground and final QA gates remain unexecuted.

Prefer verified British manufacture when considering UK machinery. A UK supplier or integrator does not establish manufacturing origin. Complete local content remains unknown until origin-traced bills and configured offers exist.

Evacuation and publication

The preferred research layout has one beam per direction without a continuous external walkway. Necessary magnetic faces, supports and connections remain. Vertical spiral chute research covers variable height, anchorage, landings, assistance, fire/weather, partial failure, rescue and driverless use. Walkway removal needs accepted route and onboard alternatives at every allowed stopping point.

Videos are final-position visibility sequences. They do not simulate motion, collision-free erection, crane operation or chute deployment. CAD integrity, PDE residuals, virtual controller ticks and consistent publication hashes each establish only their checked digital scope. Closure work and v0.5.2 development notes remain explicit.

osr-maglev-control

crates/osr-maglev-control/README.md

Portable station-powered ferrite-maglev study core, using Rust `no_std`, no heap and no external dependencies. It targets simulation and off-the-shelf SBC replay. Station LSM demand is separate from onboard active PM support and guidance. Every computed output has `physical_output_authorised = false`; the I/O boundary records study outputs and supplies no current, gate-drive or PWM operation.

The existing `evaluate(previous, inputs, now_ns, params)` API remains a single-module integer current hypothesis. Its polarity, damping, envelope, time/replay, stop/reset and support-power tests remain compatible. Increasing an upward-attraction gap asks for increasing support. The controller cannot be reused unchanged for shaped-pole passive vertical support.

24-module force and moment study

`array::evaluate_array` adds coupled heave, roll, pitch, lateral displacement and yaw feedback and `array::allocate` distributes requested forces and moments over 24 support stations. This is one rigid car; the complete moving six-car Baghdad train and a validated six-degree-of-freedom suspension remain open. The array model has no longitudinal actuator; nonzero longitudinal demand appears as an unachieved residual. The separate station module supplies an offline LSM demand screen, without closing physical propulsion/braking, traction/support coupling, flexible vehicle/beam modes, finite-angle dynamics or contact behavior.

Coordinates use x forward, y left and z up, with the assumed car centre of mass as origin. Station positions and gaps are metres, force is newtons, moments are newton-metres, small angles are radians and timestamps are monotonic nanoseconds. Upward force is positive attraction. Lateral force is positive left. Each moment is the sum of `position cross force`. The explicit seed geometry is 12 paired stations at x = -5.5, -4.5, …, 5.5 m and y = ±0.88 m, z = 0 m; this matches the candidate 12 m magnetic strips. Actual centre of mass, attachment heights, lever arms, tolerances and dynamic mass properties need physical closure.

Each station has a bounded upward channel and a signed lateral channel. The illustrative limits are 0–20 kN upward, ±4 kN lateral and 2 MN/s slew per channel. These numbers are hypothetical force envelopes, not measured magnetic maps, current/voltage/thermal limits or proved simultaneous lift/guidance capability. A real PM-biased module can retain force after drive loss: a failed station takes an explicit specified residual force. Zero in the example is an ideal failure assumption. Unknown failed-channel forces require a separate bounded uncertainty and protection model before hardware use.

The allocator uses at most 256 bounded coordinate sweeps to minimise squared force residuals and moment residuals scaled by a declared 6 m lever arm. It reports the independently reconstructed resultant, residual wrench, constraint limits and feasibility tolerances. A solver iteration count establishes no physical deadline. Infeasible requests remain visible; no six-axis success is inferred from solver completion. Slew limits apply against a prior force frame; without one the first study allocation has no force history and applies only static limits. This is not a physical startup or touchdown procedure.

The mode estimator fits a small-angle rigid plane to 24 dual-gap samples and fits lateral displacement/yaw to 24 dual lateral samples. Every individual gap and lateral sample is checked before averaging; disagreement, nonfinite values and degenerate geometry fail. This fit is not a flexible-body observer. Sensor channels share one supplied acquisition timestamp; per-channel skew, physical independence, common-cause failure and calibration are not established. Derivatives use acquisition time. Host time/sequence checks reject stale, future, replayed, regressing and late frames.

A stop request preserves fresh bounded support. A station failure, active bound or infeasible allocation latches a traction stop while available ideal support remains active. Missing valid sensing or power requests external local fallback, never frozen study forces. A prior fault inhibits traction until a fresh healthy frame, reported stopped/secured conditions and reset coincide. Capture, fallback, independent protection energy and secure-condition verification need separate engineering; they are not supplied by these Boolean reports.

Reproducible offline checks

Run:

```
cargo test -p osr-maglev-control
cargo clippy -p osr-maglev-control --all-targets -- -D warnings
cargo run -p osr-maglev-control --example sbc_replay
cargo run -p osr-maglev-control --example multimodule_replay
```

The array replay integrates ten seconds of virtual 1 ms frames, prints CSV at 500 ms intervals and asserts feasibility and a final small five-mode error. Its 27,750 kg AW2 mass is a retained numerical comparison: 18,750 kg tare plus 120 × 75 kg passengers. The current station-powered design instead targets 14,200 kg tare; this retained control case is not the new

vehicle mass closure. Gains, diagonal inertias, force limits and instantaneous actuator response are synthetic. It starts with coupled displacement/tilt, commands a stop at 0.5 s and loses station 0 at 1 s under a specified zero-residual failure hypothesis. It keeps fresh support and the stop latch active. This trajectory is a numerical example, not a closed-loop stability or ride-quality proof.

`ideal::IdealPlant` integrates a constant wrench exactly within each small-angle step. Tests independently check uniform equilibrium (272,227.5 N weight), moment signs, force/moment reconstruction, five-component demand, failed-force redistribution, capacity and slew infeasibility, all-failed residuals, sensor and time rejection, support-preserving stops and secure reset. Mechanical interface power agrees with resultant wrench power, and one-step work agrees with change in kinetic plus gravitational energy. These conservation checks validate the ideal mechanics arithmetic, not actual magnet energy or electrical losses.

`io::StudyIo`, `host_tick` and `ReplayIo` define an acquisition/record boundary without owning an OS clock or hardware. Recording/acquisition failures return to the host; the host must engineer its local fallback. Repeated single-frame stub reads are rejected as replay. No scheduler, GPIO, ADC or network transport is validated, and this example measures neither SBC timing nor I/O latency.

Physical gates remain open: calibrated nonlinear magnetic force/loss maps over gap/current/temperature and shared flux, identified electrical/thermal and mechanical dynamics, actuator failure residuals, independent sensors/protection, measured end-to-end target timing, multi-module rigs, flexible guideway and loaded vehicle tests, braking/capture and train/route acceptance. No operational or manufacturing authority is inherited from the numerical study.

Flexible guideway numerical demonstrator

`coupled_bridge` connects this exact array core to the separately labelled Python beam demonstrator in `engineering/ferrite_maglev/coupled.py`. From the repository root, run:

```
cargo build -p osr-maglev-control --release --example coupled_bridge
.venv/bin/python -m engineering.ferrite_maglev.coupled --duration-s 3
.venv/bin/python -m pytest -q engineering/analysis/tests/test_ferrite_maglev_coupled.py
```

The Python host sends all 24 local relative gaps each virtual millisecond. Rust returns bounded support forces, which load a 25 m simply supported Euler–Bernoulli beam; its local displacement feeds the next sensor frame. This is two-way offline coupling, with no physical output. The beam assumes the current configured section and native quantity-derived line mass, with 2% modal damping, retaining six vertical modes of a 50-element consistent-mass model. Independent point-load closed forms check static response; analytical pinned-beam frequencies and virtual-work/undamped-energy checks control the numerical implementation.

The model can show global beam displacement exceeding 8 mm while the car follows that displacement and the relative gaps remain inside the example's study envelope. This establishes neither a gap requirement nor structural acceptance. It omits dead-load camber, torsion/corbel flexibility, moving/train loads, flexible car structure, soil, joints, fatigue, real magnetic/electrical dynamics and independent protection. The line protocol accepts at most 20,000 frames, 8 KiB per frame, and measures no hardware timing. The three-mode car motion and vertical beam coupling do not provide a six-degree-of-freedom stability proof.

Offline evacuation sequencing

`egress::evaluate` adds allocation-free doorway sequencing with an exact declared candidate/door context, fresh monotonic frames, explicit readiness/assistance prerequisites, latched contradiction/context/deadline faults, and secured/stowed reset. It reports study traction inhibition and continued-support obligations; it supplies no door, restraint, chute or support output adapter. Input assertions are unauthenticated study facts, not verified independent sensor or safety evidence.

`cargo run -p osr-maglev-control --release --example egress_replay -- REVISION PATH_CLEAR HEIGHT_COVERED` prints a bounded nine-frame synthetic sequence. The linked Python study supplies its candidate tag and calculated hypothetical path/height gates, and retains negative controls. Loss of prerequisites after deployment begins holds the sequence; no automatic retraction is commanded. The egress/support composition test preserves fresh support control while inhibiting traction. Physical timing, independent protection, accessible passenger evacuation and railway approval remain open.

Station launch and receiving study

`station` evaluates bounded offline launch, coast, receive and secured modes. It checks exact configuration/zone identity, fresh monotonic frames, phase uncertainty, reservation assertions, magnetic-band overlap and assumed current, voltage and thermal limits. Coast demands zero station thrust. Receiving braking requires the declared brake resistor to be ready. Receiver loss after commitment requires independent stopping/recovery; clearing a Boolean drive demand does not stop an already coasting train.

```
cargo run --locked -p osr-maglev-control --example station_replay
./osr design protocol
```

The scalar force/current relation, winding R/L/flux and voltage/loss calculations are explicit hypotheses. They are not measured motor maps, three-phase hardware commutation, shared support/thrust qualification or a six-car electromagnetic simulation. An external `launch_committed` assertion is not an authenticated interlocking or detector record. Persistent corridor/berth ownership is exercised separately in `osr-interlocking`; permission expiry cannot release a committed train, and full-consist clearance is required. No physical output adapter or railway operating authority is supplied.

Adaptive research tuning

`./osr design iterate` varies bounded proportional and damping gains alongside vehicle mass allocations and guidebeam sections/spans. The bridge accepts mass and gain scales as optional command arguments; nonfinite or out-of-bound scales are rejected before processing frames. Exact revisions face separate confirmation scenarios and replay/excess-gap negative controls. This offline tuning establishes neither measured magnetic response nor stability over the full operating envelope.

OpenSourceRail architecture

docs/ARCHITECTURE.md

OpenSourceRail v0.5.2 development uses station-powered ferrite maglev. Fixed station linear synchronous motors launch and receive each train. Interstation spans have passive magnetic reaction iron and no propulsion supply. Onboard ferrite PM-biased active support and guidance remain powered independently, alongside passenger services, protection and emergency equipment.

The objective is broad city access with compact elevated guidebeams, fewer moving running-gear parts and practical local manufacture. A 99% elevated-mainline planning target still depends on route rights, utilities, ground, curvature, station access, evacuation and recovery.

Configuration and interfaces

Baghdad uses six 17 m bodies. The 104.5 m coupled footprint is an assumption until couplers and gangways are resolved. The central study targets 14.2 t tare/car, including a 5.2 t equipped GFRP sandwich body, 3 t ferrite, 30 kWh services storage and 1.8 t growth reserve. Four doors per car are the body-study basis. No complete weighed body, load-bearing joint set or qualified magnetic circuit establishes these mass targets.

The ordinary civil candidate is a 25 m, 0.80 × 1.80 m box with 100 mm webs, 150 mm flanges and end infills. Hollow columns and caps are explicit quantity studies. Shallow, single-shaft and pile-group foundations remain alternative families, selected only with ground and structural evidence. Reaction iron, anchors, bearings, joints, rail survey and all construction stages remain in scope.

Layer	Controlled definition	Implemented evidence and boundary
Service and city	Geography, demand, formation, runtime, headway and station spacing	Planning packages; flat reference is 90 s/km and 190 s headway, with grade/coasting and arrival gates
Mechanical	Structural sandwich panels, four doors, ferrite cartridges, support/guidance and services	Quantity takeoffs and partial solids; complete hardpoints, joints, mass/CG/inertia and qualification open
Station propulsion	LSM cassettes/groups, phase sensing, bridges, DC supply, receiving braking and charging	Jerk-limited demand, actual assumed magnetic-band overlap and target cost cases; force/current/voltage maps unqualified
Civil	Guidebeams, supports, foundations, stations and interfaces	Exact section/end-infill quantities, gross elastic screens and foundation sensitivities; no accepted strength or soil capacity
Software	Independent support control and persistent launch/corridor/berth ownership	Offline Rust tests/replays; no qualified sensing, physical actuator adapter or measured hardware timing
Industrialisation	Moulds, cure/release, crews, transport, cranes, joints and survey	Conditional bottleneck capacity; no measured production or supplier-price evidence
Acceptance	Exact configuration, source, unit/serial identity, tests and reviewers	Source-bound receipts; engineering/manufacturing/physical acceptance remains open

Infrastructure policy, configuration, operating study and structure study control the interfaces. City `city.json` inputs feed the same current compiler.

Launch, receiving and recovery

A launch reserves the interstation corridor, receiving berth and protective boundary for one train. Permission expiry does not release committed occupancy. Release requires complete-consist clearance evidence; lost communications or a timer cannot establish that clearance. The receiver must remain able to stop a train after departure. Independent stopping/holding, receiver faults, low-service energy and stranded-train recovery require their own engineered designs.

Coasting feasibility depends on grade, drag, wind, launch speed and arrival margin. An unpowered span cannot restart a stationary train from a distant station. Terminal return geometry, depot movement and rescue equipment remain explicit requirements. The nominal trajectory does not close these obligations.

At the hot-services assumption, 45 s charging within a 60 s dwell requires 1.20 MW/train before charger losses when the battery serves the remaining dwell. Station power/charging is a separate priced target from the incomplete civil subtotal. Complete network economics must include both, plus stations, fleet, recovery, land, qualification, factory capital and financing.

Evidence boundary

Ring-field controls and block orientation seeds do not demonstrate railway lift, thrust or magnet-mass gains. Gross beam sag is distinct from relative magnetic-gap error. The coupled demonstrator remains one stationary rigid car with ideal forces and a flexible vertical beam; moving six-car dynamics, flexible bodies, torsion, joints, soil and calibrated electromagnetic/thermal response remain open.

Native CAD covers partial assemblies. Visibility videos show final-position parts and an illustrative sequence; they do not establish collision-free motion, erection or chute deployment. Continuous-walkway removal remains conditional on accepted onboard and route evacuation, including assisted access and safe landings at every permitted stop. All release flags remain false. Accepted engineering and physical releases remain zero.

Roadmap

docs/ROADMAP.md

OpenSourceRail v0.5.2 development coordinates station-powered ferrite maglev across mechanical, civil, software, manufacturing and city planning. Station LSMs launch and receive the train; independent onboard support and passenger services remain active during the unpowered coast. Implemented work and the current report identify the tested digital scope. Accepted engineering and physical releases remain zero.

Primary maglev closure queue

Package	Evidence required for closure
Requirements and configuration	Approved six-car loads, service/grade envelope, door layout, tolerances and quantitative acceptance criteria
GFRP body	Sized orthotropic panels, longitudinal members, magnetic hardpoints, joints, openings and coupler/crash paths; fire, heat, fatigue, repair and weighed equipment closure
Magnetic assemblies	Complete shared or separately priced support/guidance/thrust circuits, cartridge retention, capture, sensors and tolerances
Magnetic force and loss	Converged finite-geometry/steel/coil maps over gap, phase, current and temperature, followed by rig correlation; no automatic ring-paper gain credit
Station propulsion	Selected fixed-group/section topology, phase registration, winding R/L/flux, current/voltage/thermal limits, receiving brakes, fault response and actual purchasing quantities
Service energy and charging	Verified hot/normal duty, battery life/reserve, complete dwell charging, power-loss support and recovery mission
Civil kit	Prestress/rebar/cover/end blocks, bearings, rail anchors, joints, hollow support connections, ground-specific foundations and every handling/construction stage
Launch and protection	Atomic corridor/receiving-berth reservation, persistent committed ownership, proved tail clearance, independent receiving readiness and stopping/holding under faults
Coupled dynamics	Moving six-car loads, flexible body, lateral/torsional response, curves, joints and soil with calibrated magnetic/electrical/thermal dynamics
Hardware	Isolated I/O, current drives, protection/watchdogs and measured end-to-end timing on selected SBC/MCU equipment
City delivery	Grade/coasting arrival margins, surveyed routes/utilities, terminal return geometry, depot/recovery access and evacuation at every stop location
Factory and erection	Measured mould/cure cycles, strength releases, delivery/crane-at-radius plans, joint completion, survey acceptance and bottleneck output
Economics and origin	Complete origin-traced MBOM, configured supplier offers, station/civil/fleet scope, capital/tooling, maintenance, financing and uncertainty
Demonstrator	Manufactured serials and exact revisions, independent inspections, measured system operation and physical acceptance

The implemented station study adds jerk-limited launch/coast/arrival calculations, magnetic-band overlap, receiving-brake duty and headway/power/cost sensitivities. The structure study adds quantity-derived GFRP cases, orientation seeds, five beam families, support/foundation screens and a two-shift bottleneck model. These are calculated candidates and production targets; they do not supply the missing structural, magnetic or measured factory evidence.

The adaptive designer uses preceding results to propose vehicle-mass, beam/span and actual Rust-gain variants. It preserves ancestry/traces and uses separate confirmation cases. Its current coupled plant remains a stationary single rigid car with ideal magnetic forces. A preferred candidate therefore remains a bounded research result, without automatic promotion to accepted geometry or city criteria.

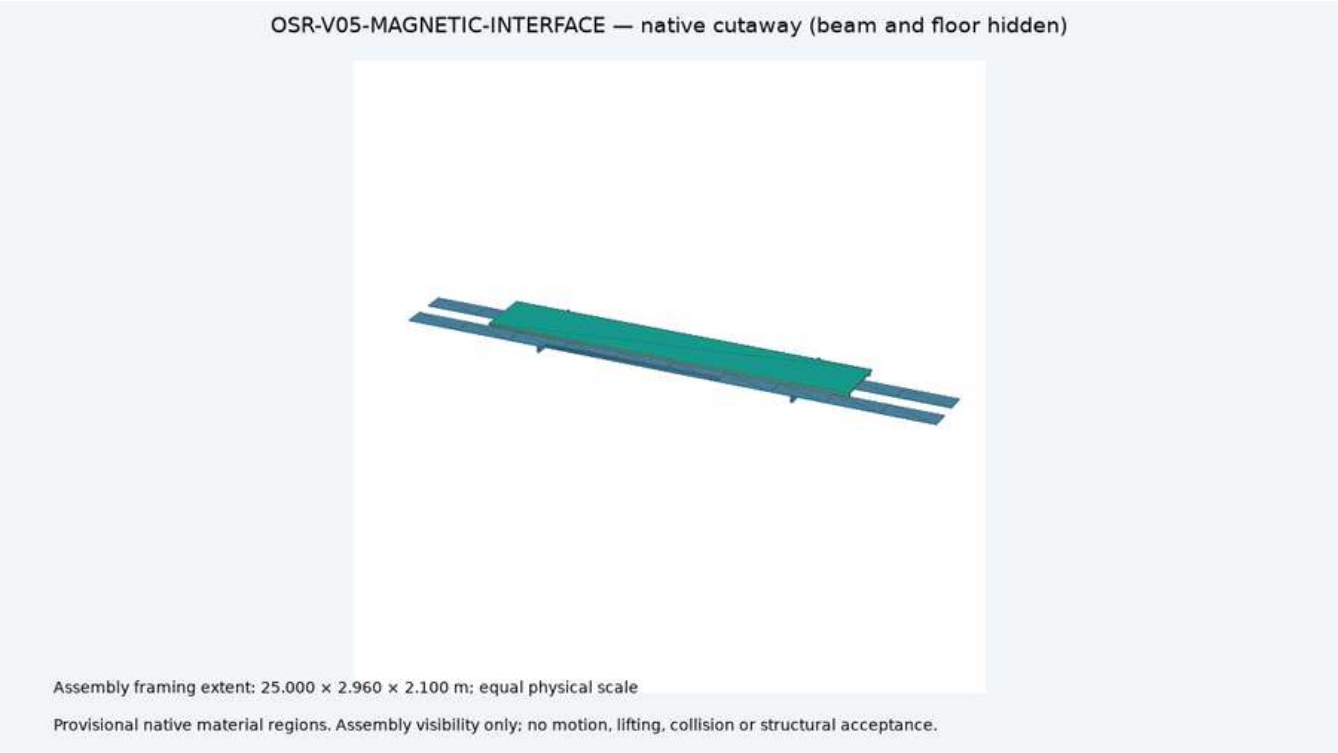
The dependency runner rebuilds stale implemented studies and holds downstream outputs after failures. Completing every automated check does not complete the engineering queue. Published v0.5.1 remains a distinct preserved release; this revision is documented in v0.5.2 development notes.

Native geometry review



Partial six-car Baghdad station, magnetic guidebeams and island lift; acceptance remains open.

Partial six-car Baghdad station, magnetic guidebeams and island lift; acceptance remains open.



Partial fixed magnetic interface; not a complete vehicle.

Partial fixed magnetic interface; not a complete vehicle.



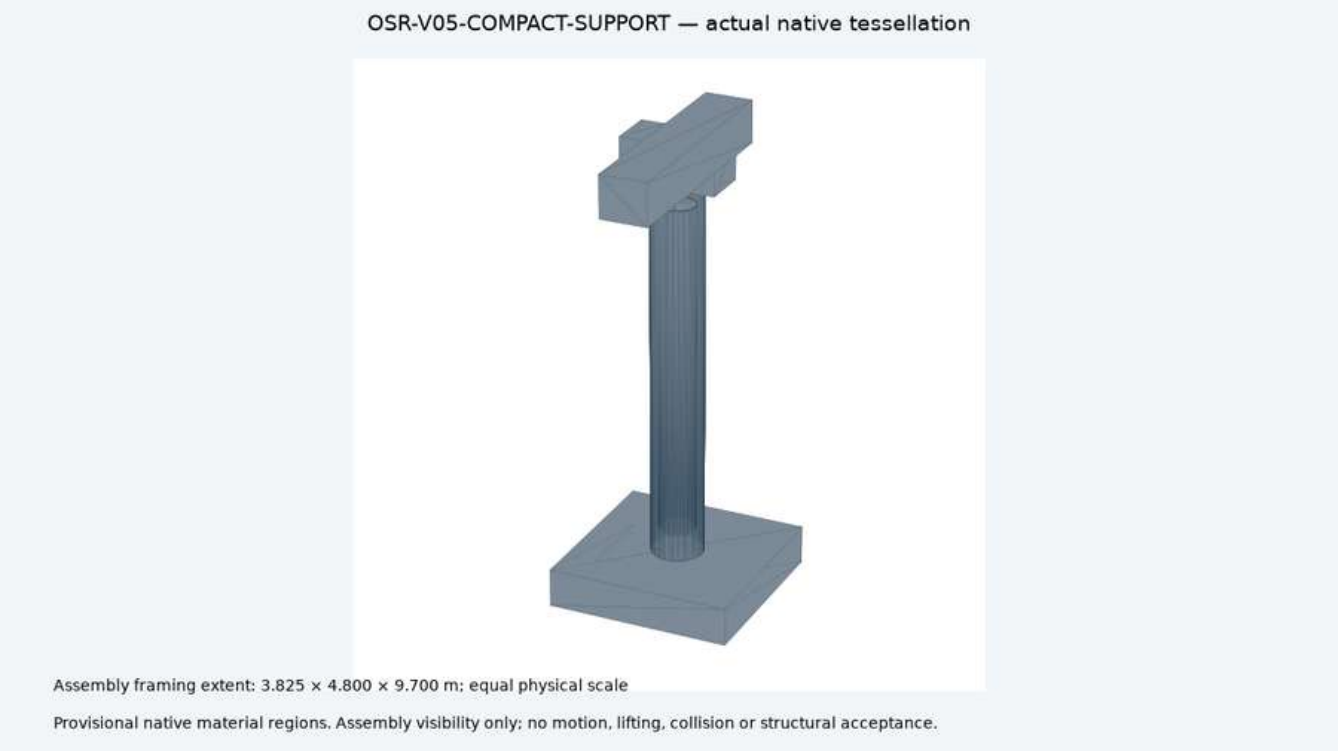
Beam-only civil variant. Supports are envelopes; evacuation acceptance remains open.

Beam-only civil variant. Supports are envelopes; evacuation acceptance remains open.



Partial evacuation entry frame and empty cassette. No deployed or stowed mechanism is qualified.

Partial evacuation entry frame and empty cassette. No deployed or stowed mechanism is qualified.



Compact pier and foundation envelope. Site capacity and structural acceptance remain open.

Compact pier and foundation envelope. Site capacity and structural acceptance remain open.

Primary maglev city directory

Separate planning packages. Complete programme costs, local content, land and physical acceptance remain open for every city.

City	Provisional trainsets	Cars	Open gates
Baghdad	268	1608	12
Benguela	70	210	12
Huambo	55	165	12
Lobito	43	129	12
Luanda	325	1950	12
Lubango	69	207	12
Malanje	18	54	12
Namibe	45	90	12
Soyo	39	78	12
Uige	12	36	12
Bukavu	64	192	12
Goma	47	141	12
Kananga	36	144	12
Kinshasa	670	4020	12
Kisangani	37	148	12
Lubumbashi	105	420	12
Mbuji Mayi	71	284	12
Eldoret	74	222	12
Garissa	31	62	12
Kakamega	36	72	12
Kisii	29	58	12
Kisumu	44	132	12
Kitale	36	72	12
Machakos	41	82	12
Malindi	37	74	12
Meru Ke	36	72	12
Mombasa	213	852	12
Nairobi	391	2346	12
Naivasha	43	86	12
Nakuru	41	123	12
Nyeri	30	60	12
Thika	50	150	12
Antananarivo	266	1596	12
Beira	36	108	12
Chimoio	24	72	12
Lichinga	17	34	12
Maputo	145	580	12
Nacala	46	92	12
Nampula	40	120	12
Pemba Mz	35	70	12
Quelimane	10	30	12
Tete	46	138	12
Xai Xai	27	54	12
Huye	38	76	12
Kigali	112	448	12
Rubavu	34	68	12
Mogadishu	141	564	12
Waw	35	70	12
Arusha	40	120	12
Dar Es Salaam	404	2424	12
Dodoma	45	135	12
Iringa	39	78	12
Kigoma	33	66	12
Mbeya	39	117	12
Morogoro	41	123	12
Moshi	30	60	12
Mwanza	131	524	12

City	Provisional trainsets	Cars	Open gates
Shinyanga	51	102	12
Songea	13	26	12
Sumbawanga	26	52	12
Tabora	32	64	12
Tanga	43	129	12
Zanzibar City	44	132	12
Arua	36	72	12
Entebbe	47	94	12
Fort Portal	32	64	12
Gulu	69	207	12
Hoima	30	60	12
Jinja	48	96	12
Kampala	160	640	12
Lira	45	90	12
Masaka	31	62	12
Mbale	37	74	12
Mbarara	43	129	12
Soroti	18	36	12
Lusaka	179	1074	12
Lyon	184	736	12
La Paz	125	500	12
Cuenca	54	162	12
San Salvador	148	592	12
Agadir	57	171	12
Beni Mellal	28	56	12
Fez	113	452	12
Kenitra	55	165	12
Khouribga	31	62	12
Marrakech	170	680	12
Meknes	43	129	12
Nador	37	74	12
Oujda	35	105	12
Safi	40	120	12
Tangier	129	516	12
Tetouan	46	138	12
El Obeid	47	141	12
Kassala	27	81	12
Khartoum	268	1608	12
Nyala	41	123	12
Omdurman	225	900	12
Port Sudan	41	123	12
Tunis	135	540	12
Bloemfontein	93	279	12
Durban	291	1746	12
East London Za	49	147	12
Nelspruit	33	66	12
Polokwane	42	126	12
Herat	37	111	12
Jalalabad Af	43	129	12
Kabul	246	1476	12
Kandahar	49	147	12
Mazar E Sharif	56	168	12
Barisal	54	162	12
Chittagong	260	1560	12
Comilla	80	240	12
Gazipur	162	648	12
Khulna	136	544	12
Mymensingh	43	129	12
Narayanganj	61	183	12
Rajshahi	66	198	12
Rangpur	39	117	12
Sylhet	41	123	12
Agra	92	368	12

City	Provisional trainsets	Cars	Open gates
Bhopal	124	496	12
Coimbatore	221	1326	12
Indore	196	1176	12
Jodhpur	102	408	12
Kanpur	183	1098	12
Lucknow	175	1050	12
Madurai	129	516	12
Meerut	79	316	12
Patna	223	892	12
Raipur	83	332	12
Rajkot	73	292	12
Ranchi	146	584	12
Vadodara	108	432	12
Varanasi	120	480	12
Vijayawada	105	420	12
Visakhapatnam	131	524	12
Biratnagar	28	56	12
Kathmandu	187	748	12
Pokhara	63	189	12
Bahawalpur	49	147	12
Faisalabad	121	726	12
Gujranwala	112	448	12
Hyderabad Pk	165	660	12
Karachi	338	2028	12
Larkana	29	87	12
Multan	102	408	12
Peshawar	108	432	12
Quetta	71	284	12
Rahim Yar Khan	55	165	12
Sheikhupura	46	138	12
Sialkot	42	126	12
Sukkur	40	120	12
Colombo	244	1464	12
Galle	52	156	12
Jaffna	49	147	12
Kandy	52	156	12
Phnom Penh	216	864	12
Bandung	226	904	12
Surabaya	238	1428	12
Vientiane	59	177	12
Mandalay	133	532	12
Yangon	351	2106	12
Davao	160	640	12
Ouagadougou	154	616	12
Bafoussam	50	150	12
Bamenda	44	132	12
Bertoua	19	57	12
Douala	126	756	12
Edea	8	16	12
Garoua	35	105	12
Kumba	34	102	12
Maroua	46	138	12
Ngaoundere	30	90	12
Yaounde	175	1050	12
Conakry	76	304	12
Bamako	225	900	12
Niamey	128	512	12
Aba Ng	32	96	12
Benin City	124	496	12
Ibadan	153	918	12
Ilorin	39	117	12
Jos	50	150	12
Kano	257	1542	12

City	Provisional trainsets	Cars	Open gates
Maiduguri	114	456	12
Onitsha	132	528	12
Port Harcourt	128	512	12
Uyo	37	111	12
Dakar	219	1314	12
Arish	14	28	12
Asyut	41	123	12
Beni Suef	32	96	12
Damanhur	50	150	12
Damietta	55	165	12
Fayoum	45	135	12
Hurghada	35	70	12
Ismailia	44	132	12
Kafr El Sheikh	37	74	12
Luxor	59	177	12
Mahalla	32	96	12
Mansoura Eg	38	114	12
Minya	47	141	12
Port Said	33	99	12
Qena	53	159	12
Sohag	46	138	12
Suez	45	135	12
Tanta	53	159	12
Zagazig	34	102	12
Amarah	40	120	12
Baqubah	51	153	12
Basra	203	1218	12
Diwaniyah	50	150	12
Duhok	42	126	12
Erbil	92	368	12
Fallujah	44	132	12
Hillah	47	141	12
Karbala	112	448	12
Kirkuk	114	456	12
Kut	37	111	12
Mosul	139	556	12
Najaf	128	512	12
Nasiriyah	41	123	12
Ramadi	50	150	12
Samawah	45	135	12
Sulaymaniyah	72	288	12
Amman	227	1362	12
Aqaba	34	68	12
Irbid	42	126	12
Zarqa	49	147	12
Beirut	116	464	12
Sidon	35	70	12
Tripoli Lb	37	111	12
Gaza City	28	84	12
Hebron	49	147	12
Nablus	53	159	12
Abha	50	150	12
Al Kharj	52	156	12
Buraidah	52	156	12
Dammam	157	628	12
Hail	58	174	12
Hofuf	51	153	12
Jeddah	244	1464	12
Jizan	46	138	12
Khamis Mushait	75	225	12
Mecca	130	520	12
Medina	115	460	12
Najran	42	126	12

City	Provisional trainsets	Cars	Open gates
Tabuk	52	156	12
Taif	48	144	12
Aleppo	195	780	12
Damascus	159	636	12
Deir Ez Zor	46	138	12
Hama	61	183	12
Homs	34	102	12
Idlib	37	74	12
Latakia	47	141	12
Raqqa	34	102	12
Tartus	33	66	12
Aden	47	141	12
Dhamar	35	70	12
Hodeidah	32	96	12
Ibb	56	168	12
Lahij	34	68	12
Mukalla	62	186	12
Sanaa	228	1368	12
Sayun	28	56	12
Taiz	45	135	12